

Moduli of VBs Talk 4.

Recall:

$$\begin{array}{ccccc}
 \{\text{semistable v.bs on } C\} / \sim_S & \longrightarrow & K(\text{Coh}(C)) & \longrightarrow & H^{2*}(C; \mathbb{Q}) \\
 & & \parallel & & \parallel \\
 & & \mathbb{Z} \oplus \text{Pic}(C) & & \mathbb{Q} \oplus \mathbb{Q}[-2] \\
 \mathcal{E} & \longmapsto & (\text{rk } \mathcal{E}, \det \mathcal{E}) & \longmapsto & (\text{rk } \mathcal{E}, \deg \mathcal{E}) \\
 \mathcal{M}_C(r, d) & \xrightarrow{\hspace{10em}} & & & (r, d) \\
 \cup \\
 \mathcal{M}_C(r, \mathbb{Z}) & \longmapsto & (r, \mathbb{Z})
 \end{array}$$

Def. (S-equivalence) S. Seshadri

$\mathcal{E} \sim_S \mathcal{E}'$ iff $\text{Gr } \mathcal{E} \cong \text{Gr } \mathcal{E}'$ w.r.t. Jordan-Hölder filtration

E.g. on EC ,

$$F_2 \sim_S \mathcal{O}_E^{\oplus 2} \not\sim_S \mathcal{O}_E(p-q) \oplus \mathcal{O}_E(q-p)$$

Rmk. (Topological classification)

$$\begin{array}{l}
 \text{Pic}(C) \xrightarrow{\deg} \mathbb{Z} \cong \text{Pic}^{\text{top}}(C) \\
 \text{Vect}_r(C) \xrightarrow{\deg} \mathbb{Z} \cong \text{Vect}_r^{\text{top}}(C)
 \end{array}$$

Top v.b. on C is determined by its degree and rank.

Moduli interpretation:

$$\begin{aligned} \underline{M}_C(r, \mathcal{L})(B) &= \{ \text{"family of semistable v.bs on } C" \} / \sim \\ &= \left\{ \mathcal{E} \in \text{Vect}(C \times B) \mid \begin{array}{l} \forall b \in B, \mathcal{E}_b \text{ is ss,} \\ \text{rk } \mathcal{E}_b = r, \\ \det \mathcal{E}_b \cong \mathcal{L} \end{array} \right\} / \sim \end{aligned}$$

Here, $\mathcal{E} \sim \mathcal{E}'$ iff $\exists \mathcal{L} \in \text{Pic}(B), \mathcal{E}' \sim_S \mathcal{E} \otimes \pi^* \mathcal{L}$.
($\pi: C \times B \rightarrow B$)

$\underline{M}_C(r, \mathcal{L})$ has a coarse moduli space, denoted by $M_C(r, \mathcal{L})$.
(deg $\mathcal{L} = 0$)

E.g. $M_E(1, \mathcal{L}) = \{ \mathcal{L} \}$

$$M_E(2, \mathcal{L}) \cong \mathbb{P}^1 \subseteq \text{Sym}^2 E$$

$$M_E(r, \mathcal{L}) \cong \mathbb{P}^{r-1} \subseteq \text{Sym}^r E$$

Facts. 1. $M_C(r, d)$ is a factorial proj variety of dim $r^2(g-1) + 1$.
(UFD, weaker than sm) (integral)
 $M_C(r, \mathcal{L})$ $(r^2-1)(g-1)$

$$\begin{aligned} 2. \text{Pic}(M_C(r, d)) &\cong \text{Pic}(\text{Pic}^d(C)) \times \mathbb{Z} & u := \gcd(r, d) \\ \text{Pic}(M_C(r, \mathcal{L})) &\cong \mathbb{Z} \cdot \mathcal{H}_{\mathcal{L}} & K_{M_C(r, \mathcal{L})} = -2u \mathcal{H}_{\mathcal{L}} \end{aligned}$$

3. Assume $g \geq 2$ and $(g, r) \neq (2, 2)$, then

$$M_C(r, d)^{\text{ns}} = \text{Sing } M_C(r, d) \text{ is of codim } \geq 2,$$

When $u=1$, $M_C(r, d)^{\text{ns}} = \emptyset$, $M_C(r, d)$ is a fine moduli space.

Other names:

$$\mathcal{M}_C(r, d) = \mathcal{M}_\sigma(v)$$

$$\mathcal{M}_C(r, \mathcal{O}_C) = \mathcal{M}_r = \text{SU}_C(r)$$

↑
[special unitary reps of $\pi_1(C)$]

"↑
[$\rho: \pi_1(C) \rightarrow \text{SU}(r)$]/~

Deformation theory result: On a stable v.b. $\mathcal{E} \in \mathcal{M}_C(r, \frac{1}{d})$,

$$T_{\mathcal{E}} \mathcal{M}_C(r, d) \cong H^1(C, \text{End}(\mathcal{E}))$$

$$T_{\mathcal{E}} \mathcal{M}_C(r, \mathcal{L}) \cong H^1(C, \text{End}^0(\mathcal{E}))$$

where

$$0 \longrightarrow \text{End}^0(\mathcal{E}) \longrightarrow \text{End}(\mathcal{E}) \xrightarrow{\text{tr}} \mathcal{O}_C \longrightarrow 0$$

Compute the dimension:

$$\begin{aligned} h^1(\text{End}(\mathcal{E})) &= r^2(g-1) + h^0(\text{End}(\mathcal{E})) \\ &= r^2(g-1) + 1 \end{aligned}$$

$$\begin{aligned} h^1(\text{End}^0(\mathcal{E})) &= (r^2-1)(g-1) + h^0(\text{End}^0(\mathcal{E})) \\ &= (r^2-1)(g-1). \end{aligned}$$

Task 1. Find an example of $E \in \mathcal{M}_C(2, 0_C)$.

E may come from some extension $\eta \in \text{Ext}^1(\mathcal{L}, \mathcal{L}^{-1})$.

$$\eta: 0 \longrightarrow \mathcal{L}^{-1} \longrightarrow E \longrightarrow \mathcal{L} \longrightarrow 0$$

If E is semistable, then $\deg \mathcal{L} \geq 0$;

if E is stable, then $\deg \mathcal{L} > 0$.

$$\text{Ext}^1(\mathcal{L}, \mathcal{L}^{-1}) \cong H^1(\mathcal{L}^{-2}) \cong H^0(\omega_C \otimes \mathcal{L}^2)$$

$$\chi(\omega_C \otimes \mathcal{L}^2) = g - 1 + 2\deg \mathcal{L}$$

$$\Rightarrow h^0(\omega_C \otimes \mathcal{L}^2) = g - 1 + 2\deg \mathcal{L} + \mathbb{1}_{\mathcal{L}^2 \cong 0_C}$$

Lemma 5.1. If η is non-trivial and $\deg \mathcal{L} = 1$,
then E is semistable.

Proof If not, then $\exists \mathcal{L}' \in \text{Pic}(C), \mathcal{L}' \hookrightarrow E$ and $\deg \mathcal{L}' > 0$.

$$\eta: 0 \longrightarrow \mathcal{L}^{-1} \longrightarrow E \longrightarrow \mathcal{L} \longrightarrow 0$$


$$\leadsto 0 \longrightarrow \underset{0}{\text{Hom}(\mathcal{L}', \mathcal{L}^{-1})} \longrightarrow \underset{0}{\text{Hom}(\mathcal{L}', E)} \longrightarrow \underset{0}{\text{Hom}(\mathcal{L}', \mathcal{L})} \longrightarrow \dots$$

$$\Rightarrow \mathcal{L}' \cong \mathcal{L}, \eta \text{ splits.}$$

The next lemma tells us that, when $g=2$, every $\mathcal{E} \in \mathcal{M}_C(2, \mathcal{O}_C)$ can be constructed in this way.

Lemma 5.5. Fix $\mathcal{E} \in \mathcal{M}_C(r, \mathcal{O}_C)$

- (i) $\forall \tilde{\mathcal{L}} \in \text{Pic}^g(C), \quad \text{hom}(\tilde{\mathcal{L}}^{-1}, \mathcal{E}) \geq r;$
- (ii) $\exists \mathcal{L} \in \text{Pic}^{g-1}(C), \text{ s.t. } \text{Hom}(\mathcal{L}^{-1}, \mathcal{E}) \neq 0;$
- (iii) The set

$$\Theta_{\mathcal{E}} := \{\mathcal{L} \in \text{Pic}^{g-1}(C) \mid \text{Hom}(\mathcal{L}^{-1}, \mathcal{E}) \neq 0\}$$

is of codim ≤ 1 ;

Moreover, when $\Theta_{\mathcal{E}}$ is a divisor, $\Theta_{\mathcal{E}} \in |r\Theta| \cong \mathbb{P}^{r^2-1}$.

Proof. (i) $\chi(\mathcal{E} \otimes \tilde{\mathcal{L}}) = r \Rightarrow h^0(\mathcal{E} \otimes \tilde{\mathcal{L}}) \geq r$

(ii) $\chi(\mathcal{E} \otimes \mathcal{L}) = 0$

Choose $\tilde{\mathcal{L}} \in \text{Pic}^g(C)$ s.t. $(\tilde{\mathcal{L}}^{-1})^{\oplus r} \neq \mathcal{E}$. (*)

Fix $\mathbb{C}^r \subseteq \text{Hom}(\tilde{\mathcal{L}}^{-1}, \mathcal{E})$.

$$\begin{array}{ccc} \tilde{\mathcal{L}}^{-1} \otimes_{\mathbb{C}} \text{Hom}(\tilde{\mathcal{L}}^{-1}, \mathcal{E}) & \longrightarrow & \mathcal{E} \\ \downarrow \cup & \searrow \phi & \\ \tilde{\mathcal{L}}^{-1} \otimes_{\mathbb{C}} \mathbb{C}^r & & \end{array}$$

(*) $\Rightarrow \phi$ is not iso

$\Rightarrow \exists p \in C$ s.t. $\phi_p: \tilde{\mathcal{L}}_p^{-1} \otimes_{\mathbb{C}} \mathbb{C}^r \longrightarrow \mathcal{E}_p$ is not inj;

i.e. $\exists \varphi \in \mathbb{C}^r \subseteq \text{Hom}(\tilde{\mathcal{L}}^{-1}, \mathcal{E})$ s.t. φ_p is not inj;

i.e. φ factor through $\tilde{\mathcal{L}}^{-1} \rightarrow \tilde{\mathcal{L}}^{-1}(p) \rightarrow \mathcal{E}$

Then $\mathcal{L} = \tilde{\mathcal{L}}(-p)$ satisfies the condition.

(iii) Idea. Writing Θ_E as the determinant loci,
and use Porteous' formula to compute $[\Theta_E] \in H^2(\text{Pic}^{g-1}(C))$

Pick $p \in C$, $N \gg 0$, $s = h^0(E(N_p))$. we have a SES

$$\begin{aligned} 0 \longrightarrow E_1 \longrightarrow E_0 \stackrel{\text{def}}{=} \mathcal{O}_C(-N_p)^{\oplus s} \longrightarrow E \longrightarrow 0 \\ 0 \longrightarrow E_1 \otimes \mathcal{L} \xrightarrow{\phi_{\mathcal{L}}} E_0 \otimes \mathcal{L} \longrightarrow E \otimes \mathcal{L} \longrightarrow 0 \\ \leadsto 0 \rightarrow H^0(E \otimes \mathcal{L}) \rightarrow H^1(E_1 \otimes \mathcal{L}) \xrightarrow{H^1(\phi_{\mathcal{L}})} H^1(E_0 \otimes \mathcal{L}) \longrightarrow H^1(E \otimes \mathcal{L}) \rightarrow 0 \end{aligned}$$

rk	$s+r$	s
deg	$(s+r)(g-1)-N$	$s(g-1)-N$
dim	N	N

Globally, this can be written as

$$\begin{array}{ccc} & & C \times \text{Pic}^d(C) \\ & \swarrow \pi_C & \searrow \pi_P \\ & C & \text{Pic}^d(C) \end{array}$$

$$0 \rightarrow R^0 \pi_{p,*}(\pi_C^* E \otimes \mathcal{P}_d) \rightarrow R^1 \pi_{p,*}(\pi_C^* E_1 \otimes \mathcal{P}_d) \xrightarrow{H^1(\phi)} R^1 \pi_{p,*}(\pi_C^* E_0 \otimes \mathcal{P}_d) \rightarrow R^1 \pi_{p,*}(\pi_C^* E \otimes \mathcal{P}_d) \rightarrow 0$$

$$\Rightarrow \Theta_E = \mathcal{M}_{N-1}(H^1(\phi)) = \{ \mathcal{L} \in \text{Pic}^{g-1}(C) \mid \text{rk } H^1(\phi_{\mathcal{L}}) \leq N-1 \}$$

is of expected codim $1 \times 1 = 1$.

If Θ_E is a divisor, then

$$\begin{aligned} [\Theta_E] &= \Delta, \left[\frac{c(R^1 \pi_{p,*}(\pi_C^* E_0 \otimes \mathcal{P}_d))}{c(R^1 \pi_{p,*}(\pi_C^* E_1 \otimes \mathcal{P}_d))} \right] \\ &= -c_1(R \pi_{p,*}(\pi_C^* E \otimes \mathcal{P}_d)) \end{aligned}$$

We need to use GRR to compute it.

$$\begin{array}{ccccc}
 C \times \text{Pic}^d(C) & \xleftrightarrow{\mathcal{P}_d = \pi_C^* \mathcal{O}_C(dp) \otimes \mathcal{P}_0} & C \times \text{Jac}(C) & \xrightarrow{\mathcal{P}_0} & A \times \text{Pic}^0(A) \\
 \downarrow \pi_C & & \downarrow \pi_P & & \downarrow \pi \\
 C & & \text{Pic}^d(C) & \xleftrightarrow{\text{twist}} & \text{Jac}(C) & \xrightarrow{\cong} & \text{Pic}^0(A) \\
 & & \mathcal{L}(dp) & \xleftrightarrow{\quad} & \mathcal{L} & &
 \end{array}$$

We know that

$$c_1(\mathcal{P}) = \sum e_i \wedge e_i^* \quad \text{[Huy 06, p198]} \quad \text{dual basis}$$

$$[\Theta] = \sum_{i=1}^g \alpha_i \wedge \beta_i$$

Write

$$\begin{aligned}
 \gamma &= c_1(\mathcal{P}_0) = \sum_i \alpha_i^C \wedge \hat{\alpha}_i^T \in H^1(C) \times H^1(\text{Jac}(C)) \\
 \eta &= \pi_C^*[p] \in H^2(C) \times H^1(\text{Pic}^d(C))
 \end{aligned}$$

symplectic basis, not the dual basis

Then

$$\begin{aligned}
 c_1(\mathcal{P}_d) &= d\eta + \gamma, & \gamma^3 = \eta^2 = \gamma\eta &= 0 \\
 \Rightarrow \text{ch}(\mathcal{P}_d) &= e^{d\eta + \gamma}
 \end{aligned}$$

$$r(d+1-g)[\text{Pic}^d(C)] - r[\mathcal{O}_C]$$

Claim: $ch(R\pi_{p,*}(\mathcal{P}_d \otimes \pi_c^* \mathcal{E})) = \cancel{r(d+1-g) - r} [\mathcal{O}_C]$

Proof. Using GRR,

$$\begin{aligned} ch(R\pi_{p,*}(\mathcal{P}_d \otimes \pi_c^* \mathcal{E})) \underbrace{td(\text{Pic}^d(C))}_1 &= \pi_{p,*} \left(ch(\mathcal{P}_d \otimes \pi_c^* \mathcal{E}) td(C \times \text{Pic}^d(C)) \right) \\ &= \pi_{p,*} \left(ch(\mathcal{P}_d) \cdot \pi_c^* ch(\mathcal{E}) \cdot \pi_c^* td(C) \right) \\ &= \pi_{p,*} \left(e^{d\eta + \gamma} \cdot r \cdot (1 + (1-g)\eta) \right) \\ &= \pi_{p,*} \left((1 + d\eta + \gamma + \frac{1}{2}\gamma^2) r (1 + (1-g)\eta) \right) \\ &= r \pi_{p,*} \left(1 + (d+1-g)\eta + \gamma + \frac{1}{2}\gamma^2 \right) \\ &= r \pi_{p,*} \left((d+1-g)\eta + \frac{1}{2}\gamma^2 \right) \\ &= r(d+1-g)[\text{Pic}^d(\mathcal{O}_C)] + \frac{1}{2} r \pi_{p,*}(\gamma^2) \end{aligned}$$

It reduces to show that $\pi_{p,*}(\gamma^2) = -2[\mathcal{O}_C]$.

$$\begin{aligned} \gamma^2 = \gamma \wedge \gamma &= 2 \sum_i (\alpha_i^c \wedge (-\beta_i^J)) \wedge (\beta_i^c \wedge \alpha_i^J) \\ &= 2 \sum_i \alpha_i^c \wedge \beta_i^c \wedge \beta_i^J \wedge \alpha_i^J \\ &= -2 \sum_i \underbrace{\alpha_i^c \wedge \beta_i^c}_{\alpha_i^J \wedge \beta_i^J} \end{aligned}$$

$$\begin{aligned} \pi_{p,*}(\gamma^2) &= -2 \sum_i \alpha_i^J \wedge \beta_i^J \\ &= -2[\mathcal{O}_C]. \end{aligned}$$

□

Q: How to describe $M_C(r, \mathcal{O}_C)$?

A: We can construct a rational map

$$\vartheta: M_C(r, \mathcal{O}_C) \dashrightarrow |r\Theta_C| \cong \mathbb{P}^{r^2-1}$$

[well-defined: $\Theta_{\mathcal{O}_C^{\oplus r}}$ is a divisor]

$$\mathcal{L}_\vartheta = \vartheta^* \mathcal{O}(1) \text{ generates } \text{Pic}^0(M_C(r, \mathcal{O}_C))$$

Rmk: When $r \geq 2$, $|r\Theta_C|$ is base-point free, $\forall \mathcal{L} \in \text{Pic}^{r^2-1}(C)$

$$\Rightarrow H_{\mathcal{L}} := \{D \in |r\Theta_C| \mid \mathcal{L} \in D\} \text{ is a hyperplane in } |r\Theta|$$

Denote

$$\Theta_{\mathcal{L}} = \vartheta^{-1} H_{\mathcal{L}} = \{E \in M_C(r, \mathcal{O}_C) \mid \text{Hom}(\mathcal{L}^{-1}, E) \neq 0\}$$

is a divisor in $|r\Theta|$

(check for some strict ss v.b. E , $\text{Hom}(\mathcal{L}^{-1}, E) \neq 0$)

then $\mathcal{L}_\vartheta = \mathcal{O}(\Theta_{\mathcal{L}})$.

Q: How does ϑ look like?

E.g. 1. When $g=1$, $\theta: M_E(r, \mathcal{O}_E) \rightarrow \mathbb{P}^{r-1}$ is an iso.

E.g. 2. When $g=r=2$, $\theta: M_C(2, \mathcal{O}_C) \rightarrow \mathbb{P}^3$ is an iso.

The non-stable locus $M_C(2, \mathcal{O}_C)^{ns} \subseteq \mathbb{P}^3$ is of form

Kummer surface

↓ blow up 16 pts

$$M_C(2, \mathcal{O}_C)^{ns} = \{L \oplus L^{-1} \in M_C(2, \mathcal{O}_C) \mid L \in \text{Pic}(C)\} \cong \text{Jac}(C) / \{\pm 1\}$$

E.g. 3. When $g=2$, $r=3$,

$\theta: M_C(3, \mathcal{O}_C) \rightarrow \mathbb{P}^8$ is a double covering,
 ramified over a sextic hypersurface,
 which is dual to a cubic hypersurface in $(\mathbb{P}^8)^*$
 $\uparrow$ Coble cubic

E.g. 4. When $g=2$, $r \geq 4$,

$\theta: M_C(r, \mathcal{O}_C) \dashrightarrow \mathbb{P}^{r^2-1}$ is generically finite,
 but not finite.

When $r \geq 8$, θ is only a rational map.

Q: How to compute $\deg \theta$? When $4 \leq r < 8$, is θ an iso?

E.g 5. When $g=3, r=2$,

$$\Theta: M_C(2, \mathcal{O}_C) \hookrightarrow \mathbb{P}^7$$

gives a quartic hypersurface Q , which is self-dual.
 $(Q^* \cong Q)$
 $\uparrow$ Coble quartic

E.g. 6. When $r=2, g \geq 3$, We have embeddings

$$\begin{array}{ccc} M_C(2, \mathcal{O}_C) & \hookrightarrow & |2\Theta_C| & \text{non-hyper} \\ M_C(2, \mathcal{O}_C)/\langle l^* \rangle & \hookrightarrow & |2\Theta_C| & \text{hyper} \end{array}$$

And $\deg M_C(2, \mathcal{O}_C) = (3g-3)! 2^g (2\pi)^{2-2g} \{(2g-2)$
 by Verlinde formula.

Q: When $g \geq 3$, C non hyperelliptic, is Θ always gen inj?